Uniqueness of the Two-Boundary Retrocausal Master Equation: a Giulini–Marolf–Aczél–Connes Theorem for the TBRME

Companion-paper architecture (rev. 2: Type II$_1$ gravity sector, Aczél clock re-route)

Alexandru Dima

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Abstract

This companion paper proves a uniqueness theorem for the Two-Boundary Retrocausal Master Equation (TBRME) of the main unification paper. Under seven explicit axioms on a densely-defined symmetric constraint operator $\widehat{C}$ on the universal Hilbert space $\mathcal{U}$ — clock covariance with the fibrewise Aczél–d'Alembert cocycle, modular covariance with respect to the *non-tracial* dual seed state $\hat{\omega}$ on the CLPW Type II$_1$ gravity sector, direct-sum locality in the branch index, TDSE flat-space limit, semiclassical-Einstein WKB limit, a pure-geometry *universality-class* condition in the sense of Kaimakkamis–Sil, and branch-uniform Part I normalization — we show $\widehat{C}$ equals $\lambda\,\widehat{\mathfrak{C}}_k$ uniquely up to (a) a single overall positive scale λ, (b) conjugation by unitaries of the $\hat{\omega}$-centralizer $\mathcal{M}_{\mathrm{geom}}{}^{\hat{\omega}}$, and (c) von Neumann self-adjoint-extension freedom where the deficiency indices of $\widehat{C}\restriction_{\Phi}$ are non-trivial and equal. The structural template is Giulini–Marolf rigging-map uniqueness (outer frame; it also sources the per-branch scale freedom that Axiom X7 collapses), with the scalar Aczél–d'Alembert classification supplying the clock factor (Kisyński–Sova operator-cosine theory certifying the generator), the Pedersen–Takesaki noncommutative Radon–Nikodym theorem supplying the gravity factor on the finite-trace Type II$_1$ sector, and Meda–Pinamonti–Siemssen semiclassical well-posedness closing the WKB leg. All external ingredients are in the ambient literature and are cited at theorem level; the contribution of this paper is assembly, an axiom independence/consistency audit, the over-determination lemma for the matter–gravity coupling, and explicit identification of the three irreducible ambiguity classes.

1 Introduction and context

Placeholder. State the main paper's TBRME $\widehat{\mathfrak{C}}_k = \widehat{K}_k - \kappa^{-1}\widehat{Q}^{(k)}[f]$, recap the five-fold reduction theorem (U1)–(U5), and motivate the uniqueness question. Stress the "modulo centralizer" status — this is not an iff classification; it is the same uniqueness quality as the Aczél–d'Alembert classification for the clock shift.

Two corrections relative to the first scaffold are structural and propagate everywhere:

- **Type II$_1$, not II$_\infty$.** With a positive clock Hamiltonian on $L^2(\mathbb{R}_+)$ and a compact region, the CLPW/DEHK crossed-product gravity sector is a Type II$_1$ factor with a *finite*, trace-normalized trace τ ($\tau(\mathbb{1}) = 1$) [10]; "II$_\infty$ with induced finite trace" is a contradiction in terms. The II$_\infty$ variant survives only on non-compact/black-hole-exterior sectors [11], treated in Rem. 4.3.
- **Aczél, not Kisyński, selects the clock shape.** Per ledger item A29, $C = B + \mathbb{1}$ does not satisfy the homogeneous operator-cosine equation; the clock-sector uniqueness is the fibrewise scalar Aczél–d'Alembert classification (main paper, App. A as corrected), with Kisyński–Sova theory certifying the reconstructed generator only. The axioms below import the corrected measurability/continuity hypotheses explicitly.

2 Standing structures and axioms

Let $\mathcal{U} = \bigoplus_{k\in\mathbb{K}_+} \mathcal{H}_{\rm clock} \otimes \mathcal{H}^{(k)}_{\rm geom} \otimes \mathcal{H}^{(k)}_{\rm matter}$, where each $\mathcal{H}^{(k)}_{\rm geom} = L^2(\mathcal{M}^{(k)}_{\rm geom}, \tau_k)$ carries the CLPW/DEHK crossed-product structure of App. F of the main paper in its corrected form: for a positive clock Hamiltonian on $L^2(\mathbb{R}_+)$ and compact region, $\mathcal{M}^{(k)}_{\rm geom}$ is a *Type II*$_1$ *factor* with normalized finite trace τ_k, $\tau_k(\mathbb{1}) = 1$ [10, 7]. Let $\hat\omega_k = \tau_k(h_k\,\cdot\,)$ be the dual state induced by the branch seed ω_k, with Radon–Nikodym density $h_k \in L^1(\mathcal{M}^{(k)}_{\rm geom}, \tau_k)_+$, h_k injective (faithfulness), $\tau_k(h_k) = 1$. Its modular automorphism group is *inner*,

$$\sigma^{\hat\omega_k}_s \;=\; \mathrm{Ad}\big(h_k^{is}\big), \tag{1}$$

by the Pedersen–Takesaki Radon–Nikodym theorem for semifinite algebras [7] (equivalently: a von Neumann algebra is semifinite iff its modular flows are inner). In particular the modular group of the *trace* is trivial, $\sigma^{\tau_k} = \mathrm{id}$; all modular-covariance statements below are therefore made with respect to $\hat\omega_k$, never τ_k.

Let $\widehat{C}$ be a densely-defined *symmetric* operator on a common invariant rigged-Hilbert dense domain $\Phi \subset \mathcal{U}$ (essential self-adjointness is *not* assumed; see ambiguity (c) and Rem. 3.2). Assume:

Axiom 2.1 ((X1) Clock covariance and Aczél–d'Alembert cocycle)**.** $[\widehat{C}, \widehat{H}_{\rm c} \otimes \mathbb{1} \otimes \mathbb{1}] = 0$ on Φ, where $\widehat{H}_{\rm c}$ is the Page–Wootters clock generator, and the clock-shift response $B(\delta) :=$ $\widehat{\Sigma}$-part of $\widehat{C}$ under finite clock shifts satisfies, on the band-limited subspace $\mathcal{H}_{[E_\star]}$, the package of the main paper's Prop. A (operator-sense d'Alembert reduction): $B(\delta)$ bounded, self-adjoint, even in δ, strongly continuous in δ on $\mathcal{D}(\widehat{H}_{\rm c}^2)$, with $B(\delta) \in \{\widehat{H}_{\rm c}\}''$ and the inhomogeneous cocycle

$$B(\delta_1+\delta_2) + B(\delta_1-\delta_2) = 2\cos(\delta_1\widehat{H}_{\rm c}/\hbar)B(\delta_2) + 2B(\delta_1),$$

together with $\mu_{\widehat{H}_{\rm c}}$-measurability of the fibrewise data $\xi \mapsto f(\xi,\delta)$ for each rational δ.

Remark 2.2 (A29 accommodation). X1 deliberately states the *inhomogeneous* cocycle plus the measurability/continuity hypotheses under which the fibrewise scalar Aczél classification is a theorem (main paper, Prop. *wdw-dalembert* and Thm. *wdw-selection*); no homogeneous operator-cosine hypothesis on $C = B + \mathbb{1}$ is made or used. If the sibling re-route adds further measurability hypotheses, they are to be absorbed into X1 verbatim.

Axiom 2.3 ((X2) Modular covariance w.r.t. the dual seed state, non-degenerate)**.** For each branch k there is a decomposition on Φ, $\widehat{C}_k = \mathbb{1} \otimes \widehat{C}^{(k)}_{\rm grav} \otimes \mathbb{1} + \widehat{C}^{(k)}_{\rm rest}$, with $\widehat{C}^{(k)}_{\rm grav}$ symmetric and affiliated to $\mathcal{M}^{(k)}_{\rm geom}$, $\widehat{C}^{(k)}_{\rm rest}$ commuting with h_k^{is} for all s, such that

$$\mathrm{Ad}\big(e^{it\widehat{C}^{(k)}_{\rm grav}}\big)\restriction_{\mathcal{M}^{(k)}_{\rm geom}} \;=\; \sigma^{\hat\omega_k}_{-\lambda_k t} \qquad \text{for some } \lambda_k > 0 \text{ and all } t \in \mathbb{R}. \tag{2}$$

Non-degeneracy: $\hat\omega_k$ is non-tracial, i.e. $h_k \notin \mathbb{C}\mathbb{1}$. (If h_k were scalar, $\sigma^{\hat\omega_k} = \mathrm{id}$ and (2) would be vacuous; the axiom then selects nothing. Non-degeneracy holds automatically whenever the seed modular Hamiltonian K_{ω_k} is non-scalar, i.e. in every physical branch.)

Axiom 2.4 ((X3) Direct-sum locality in the branch index)**.** $\widehat{C} = \bigoplus_{k\in\mathbb{K}_+} \widehat{C}_k$, with each $\widehat{C}_k$ affiliated to $\mathcal{B}(\mathcal{H}_{\rm clock}) \otimes \mathcal{M}^{(k)}_{\rm geom} \otimes \mathcal{B}(\mathcal{H}^{(k)}_{\rm matter})$, where $\mathcal{M}^{(k)}_{\rm geom}$ is the Type II$_1$ crossed-product factor above.

Axiom 2.5 ((X4) TDSE flat-space limit, projective form)**.** In the flat, classical-clock, finite-matter limit (reduction (U1)/(U2) of the main paper), $\widehat{C}_k \downarrow c_k\big(\widehat{H}_{\rm clock} + \widehat{H}^{(k)}_{\rm matter}\big)$ for some $c_k > 0$, strongly on the limit domain. *(Projective form: only the ray of the limit generator is fixed. The strict form $c_k = 1$ would force $\lambda = 1$ in Thm. 3.1 and render ambiguity (a) vacuous; see Sec. 7.)* The limit additionally supplies the TDSE normalization $-B''(0) = \alpha\widehat{H}_{\rm c}^2/\hbar^2$ on $\mathcal{D}(\widehat{H}_{\rm c}^2)$ used to fix the common fibrewise scale in the clock sector.

Axiom 2.6 ((X5) Semiclassical-Einstein WKB limit)**.** In the WKB-geometry sector, $\langle\widehat{C}\rangle$-vanishing yields the semiclassical Einstein equation $G_{\mu\nu} = (8\pi G/c^4)\langle T_{\mu\nu}\rangle$ (reduction (U3) of the main paper), within the scope where Meda–Pinamonti–Siemssen local existence–uniqueness holds [8]: *flat* FLRW backgrounds driven by a massive scalar with arbitrary curvature coupling, initial data at finite time, regular compatible state. (The earlier "compact-Cauchy FLRW" scope and the "Pinamonti–Drago–Hack" attribution were incorrect and are withdrawn.)

Axiom 2.7 ((X6) Pure-geometry universality class (weakened))**.** Tracing out $\mathcal{H}_{\text{clock}} \otimes \mathcal{H}^{(k)}_{\text{matter}}$ reduces $\widehat{C}_k$ to a quantum Hamiltonian constraint whose (i) classical limit is the ADM/Wheeler–DeWitt constraint of general relativity, and whose (ii) operator-ordering ambiguity function lies in the *same Kaimakkamis–Sil universality class* [9] as the minimal (Laplace–Beltrami-covariant) ordering: under the field-redefinition-invariant reformulation, all ordering ambiguities consolidate into a single higher-order scalar function on minisuperspace, and X6 requires only membership of the minimal class. *X6 makes no reference to the cosine deformation $\widehat{\Sigma}_\delta$ or to any other structure that Thm. 3.1 is supposed to deliver; the Σ_δ term is derived from X1+X4, not assumed here.*

Axiom 2.8 ((X7) Branch-uniform Part I normalization)**.** On finite-dimensional matter factors, $\widehat{C}_k$ restricts to the Part I two-boundary constraint (reduction (U2) of the main paper) *with a branch-independent normalization*: the constants c_k of X4 and the flow rescalings λ_k of X2 satisfy $c_k = c$, $\lambda_k = \lambda$ for all $k \in \mathbb{K}_+$. Equivalently: the Giulini–Marolf per-sector scale freedom [2] is fixed to a single global scale by Part I consistency of the branch weights $w(k)$.

Remark 2.9 (What X7 is for). Giulini–Marolf uniqueness is *per superselection sector*: on a direct sum the rigging map (and with it the normalization of the constraint) is unique only up to one positive constant per sector [2]. Under X3 every branch k is a superselection sector, so X1–X6 alone can at best give $\widehat{C}_k = \lambda_k\widehat{\mathfrak{C}}_k$ with an uncontrolled sequence (λ_k). X7's entire job is to collapse $(\lambda_k) \to \lambda$; it is *not* an independent rigidity input on the operator form (that part of the old X7 was implied by X4 and has been removed — see Sec. 7).

3 Main theorem

Theorem 3.1 (TBRME uniqueness, sharp form)**.** *Under Ax. 2.1–2.8 on the universal Hilbert space $\mathcal{U}$ of the main paper,*

$$\widehat{C} \;=\; \lambda \bigoplus_{k\in\mathbb{K}_+} \widehat{\mathfrak{C}}_k \qquad \textit{for some } \lambda \in \mathbb{R}_{>0},$$

uniquely up to:

(a) *the overall positive scalar $\lambda > 0$ (single and global, by X7; per-branch scales λ_k if X7 is dropped);*

(b) *conjugation by unitaries of the $\hat{\omega}_k$-centralizers $\mathcal{M}^{(k)}_{\text{geom}}{}^{\hat{\omega}} = \{h_k\}' \cap \mathcal{M}^{(k)}_{\text{geom}}$ (a genuine gauge class: such conjugations fix the clock and gravity generators but act non-trivially on the gravity–matter coupling ordering);*

(c) *a von Neumann self-adjoint-extension choice when the deficiency indices (n_+, n_-) of $\widehat{C}\restriction_\Phi$ are non-trivial and equal [12].* (Attribution corrected: this freedom is von Neumann extension theory, not Giulini–Marolf [2], whose theorem concerns rigging-map scale per superselection sector and contains no deficiency-index content.)

Remark 3.2 (Essential self-adjointness vs. ambiguity (c)). The first scaffold assumed $\widehat{C}$ essentially self-adjoint *and* kept ambiguity (c); that conjunction is inconsistent (essential self-adjointness means $(n_+, n_-) = (0, 0)$). The standing assumption is now symmetry on Φ only. In the regime where the main paper's Kato–Rellich argument (T4)+(T7) applies, $\widehat{C}\restriction_\Phi$ is essentially self-adjoint and (c) collapses; (c) is live precisely outside that regime (e.g. at minisuperspace boundaries $a \to 0$), see Sec. 5.

4 Proof architecture

4.1 Clock sector: fibrewise Aczél–d'Alembert closure

X1 (covariance + spectral theorem) forces $B(\delta) = f(\widehat{H}_c, \delta)$; the cocycle read on each spectral fibre ξ is the *scalar* inhomogeneous d'Alembert equation, whose continuous even solutions with $f(\xi, 0) = 0$ are $f(\xi, \delta) = c(\xi)(\cos(\delta\xi/\hbar) - 1)$ (Aczél [5]); measurability of $c(\xi)$ from X1's measurability clause; the TDSE normalization inside X4 forces $c(\xi) \equiv \alpha$. Kisyński–Sova operator-cosine theory [3, 4] enters only to certify that the normalized member is the cosine function of $A = (\widehat{H}_c/\hbar)^2 \succeq 0$. *Import from App. A of the main paper as corrected under A29; do not import the withdrawn $C = B + \mathbb{1}$ homogeneous-cosine step.* **[Status: proven]** on $\mathcal{H}_{[E_\star]}$; extension to unbounded $\widehat{H}_c$ without band-limit: **[Status: genuinely open]** (tedious spectral-limit argument, expected routine).

4.2 Gravity sector: Pedersen–Takesaki rigidity on the Type II$_1$ factor

The Takesaki–Connes cocycle-derivative theorem [6, Thm. VIII.3.3] is type-agnostic and survives the reclassification; but on a II$_1$ factor it must be deployed against the non-tracial state $\hat{\omega}_k$, since $\sigma^\tau = \mathrm{id}$. On the finite-trace sector the argument in fact simplifies to the following elementary lemma.

Lemma 4.1 (Gravity-sector rigidity, II$_1$ form)**.** *Let $\mathcal{M}$ be a II$_1$ factor with normalized trace τ, and $\hat{\omega} = \tau(h\,\cdot\,)$ a faithful normal non-tracial state, $h \in L^1(\mathcal{M}, \tau)_+$ injective, $\tau(h) = 1$. Suppose $\widehat{C}_{\mathrm{grav}}$ is self-adjoint, affiliated to $\mathcal{M}$, and satisfies* (2)*: $\mathrm{Ad}(e^{it\widehat{C}_{\mathrm{grav}}}) \restriction_{\mathcal{M}} = \sigma^{\hat{\omega}}_{-\lambda t} = \mathrm{Ad}(h^{-i\lambda t})$. Then*

$$\widehat{C}_{\mathrm{grav}} \;=\; \lambda\,(-\log h) \;+\; \mu\mathbb{1}, \qquad \mu \in \mathbb{R},$$

and $\mu = 0$ iff $\tau\big(e^{-\widehat{C}_{\mathrm{grav}}/\lambda}\big) = 1$. In particular the II$_1$ trace normalization $\tau(\mathbb{1}) = 1$, $\tau(h) = 1$ canonically fixes the additive constant*, a strengthening unavailable in the II$_\infty$ setting.*

Proof sketch. Set $u_t := e^{it\widehat{C}_{\mathrm{grav}}}\, h^{i\lambda t}$. By hypothesis $\mathrm{Ad}(u_t) = \mathrm{id}$ on $\mathcal{M}$, so $u_t \in \mathcal{M}' \cap \mathcal{M} = \mathbb{C}\mathbb{1}$ ($\mathcal{M}$ a factor; $u_t \in \mathcal{M}$ since both factors are affiliated to $\mathcal{M}$). Thus $e^{it\widehat{C}_{\mathrm{grav}}} = c(t)\, h^{-i\lambda t}$ with $c : \mathbb{R} \to \mathbb{T}$; strong continuity and the group law give $c(t) = e^{i\mu t}$, and Stone's theorem yields $\widehat{C}_{\mathrm{grav}} = -\lambda \log h + \mu\mathbb{1}$ on the common core. The normalization claim is functional calculus plus $\tau(h) = 1$. □

[Status: provable now, proof to be written] (the sketch is complete modulo domain bookkeeping for the affiliated logarithm, which is standard in $L^1(\mathcal{M}, \tau)$).

Remark 4.2 (Where the centralizer ambiguity (b) actually lives)*.* Lemma 4.1 pins $\widehat{C}^{(k)}_{\mathrm{grav}}$ *exactly* (up to λ_k); conjugation by a unitary $u \in \mathcal{M}^{(k)}_{\mathrm{geom}}{}^{\hat{\omega}}$ fixes h_k (hence $\widehat{C}^{(k)}_{\mathrm{grav}}$) and fixes $\sigma^{\hat{\omega}_k}$, but acts non-trivially on the gravity–matter coupling term $\widehat{Q}^{(k)}[f]$. Ambiguity (b) is therefore *not* a residual freedom of the gravity generator (as the first scaffold suggested); it is a residual gauge action on the *coupling ordering.* In a II$_1$ factor the centralizer $\{h\}' \cap \mathcal{M}$ is typically large, so (b) is a genuinely non-trivial class; partial collapse under X5 is discussed in Sec. 4.3, step (v).

Remark 4.3 (II$_\infty$ variant)*.* On non-compact sectors (black-hole exterior, CPW [11]) $\mathcal{M}$ is II$_\infty$ with semifinite trace; Lemma 4.1 survives verbatim except that the density h is τ-measurable with possibly $\tau(h) = \infty$-adjacent normalization, and the additive constant μ is no longer canonically fixed. The theorem statement is uniform: "Type II factor with faithful normal semifinite trace, finite (and normalized) in the positive-clock/compact case."

4.3 Matter sector: the over-determination lemma

After Secs. 4.1–4.2 and X3, the residual freedom in $\widehat{C}_k$ is: the pure-matter part, the gravity–matter coupling ordering in $\widehat{Q}^{(k)}[f]$, and centralizer conjugation (b).

Lemma 4.4 (Over-determination of the coupling ordering)**.** *Let $\widehat{C}, \widehat{C}'$ both satisfy X1–X7 and define $D_k := \widehat{C}_k - \widehat{C}'_k$ on Φ. Then:*

(i) (Clock.) *The clock parts of $\widehat{C}_k, \widehat{C}'_k$ coincide up to the global scale; D_k commutes with the clock algebra.* ***[Status: proven]*** *(Sec. 4.1).*

(ii) (Gravity.) *The gravity parts differ by $(\lambda - \lambda')(-\log h_k)$ only; no centralizer term survives in the generator itself.* ***[Status: provable now, proof to be written]*** *(Lemma 4.1).*

(iii) (Pure matter.) *X4 (projective TDSE limit) forces the pure-matter parts onto the same ray: the flat-limit reduction of D_k is $(c_k - c'_k)(\widehat{H}_{\text{clock}} + \widehat{H}^{(k)}_{\text{matter}})$, i.e. scale only.* ***[Status: provable now, proof to be written]*** *(strong-limit bookkeeping on the limit domain).*

(iv) (WKB order.) *X5 forces the leading WKB symbol of the coupling part of D_k to vanish: both constraints reduce to semiclassical Einstein flows through the* same *initial data, and Meda–Pinamonti–Siemssen uniqueness [8] of the semiclassical solution forces the two flows — hence the two leading symbols — to coincide, fixing in particular the coupling constant κ^{-1} relative to the matter normalization of (iii).* ***[Status: partially proven]*** *(MPS supply forward well-posedness as a theorem; the "equal limits + unique flow ⇒ equal leading symbols" step is an inverse-direction argument that must be written; expected provable on the MPS scope, i.e. flat FLRW + massive scalar).*

(v) (Ordering class.) *X6 places the pure-geometry reductions of both constraints in the Kaimakkamis–Sil minimal universality class [9]; the within-class freedom is, by their field-redefinition- invariant reformulation, a change of inner-product measure / field redefinition, i.e. unitarily implemented gauge — and the unitaries in question that preserve X2 lie in the centralizer, so the within-class freedom is* absorbed into ambiguity (b). ***[Status: partially proven]****: proven by [9] for minisuperspace models (1-dim and n-dim with scalar matter); the lift to the midisuperspace/QFT matter content of the main paper is* ***[Status: genuinely open]*** *(this is open problem O1, the principal conjectural input of the companion paper).*

(vi) (Residual = extension data.) *What survives (i)–(v) is supported on minisuperspace boundary terms (e.g. $a \to 0$): symmetric, annihilated by all interior limits, hence localized in the deficiency subspaces of $\widehat{C}_k \restriction_\Phi$. This is exactly the von Neumann extension freedom, ambiguity (c).* ***[Status: genuinely open]*** *(conjecture O2: "interior-trivial ⇒ boundary-supported" needs a precise boundary-triple formulation; plausible via standard boundary-triple theory on the DeWitt-metric minisuperspace, but not yet written).*

Consequently $\widehat{C}_k = \lambda_k \widehat{\mathfrak{C}}_k$ modulo (b) and (c), with (λ_k) per-branch.

4.4 Assembly

X3 assembles the branch constraints as a direct sum. Each branch is a superselection sector; Giulini–Marolf [2] uniqueness of the rigging map holds per sector up to a positive scale, so the assembly step alone yields $(\lambda_k)_{k \in \mathbb{K}_+}$. X7 collapses $\lambda_k \equiv \lambda$, which is ambiguity (a). **[Status: provable now, proof to be written]** (bookkeeping once (i)–(vi) are in place).

5 Self-adjoint-extension discussion (ambiguity (c))

Spell out when deficiency indices can be non-trivial for $\widehat{C} \restriction_\Phi$: in the Kato–Rellich regime of the main paper's (T4)+(T7) the constraint is essentially self-adjoint and (c) collapses; candidate non-trivial cases are minisuperspace boundaries ($a \to 0$) and non-compact matter sectors. Physical principles that collapse (c): positive clock spectrum, positivity of the semiclassical energy, DeWitt boundary condition. *Cite von Neumann extension theory [12, Thm. X.2] and, for the* *constraint-quantization frame, Marolf's single-constraint RAQ [2]; the former "Giulini–Marolf case (iii)" citation is withdrawn — no such enumeration exists in gr-qc/9902045.*

6 Relation to the main paper's App. A

The clock-shift cocycle $\widehat{\Sigma}_\delta = \cos(\delta \widehat{H}_c/\hbar) - \mathbb{1}$ of the main paper's clock-sector uniqueness (App. A, as corrected under A29: fibrewise Aczél selection, Kisyński–Sova certification) is the clock-sector projection of Thm. 3.1. The present paper extends that closure from the clock factor to the full universal Hilbert space $\mathcal{U}$.

7 Axiom audit: dependency, independence, consistency

Dependency diagram

$$
\begin{array}{rcl}
\text{X1 (cocycle + measurability)} & \xrightarrow{\text{Aczél fibrewise}} & \text{clock shape } \alpha\widehat{\Sigma}_\delta \\
\text{X4 (TDSE, projective)} & \xrightarrow{\text{normalization}} & \text{fibre scale } c(\xi) \equiv \alpha\text{; pure-matter ray} \\
\text{X2 } (\hat{\omega}\text{, non-tracial)} & \xrightarrow{\text{Pedersen–Takesaki}} & \widehat{C}_{\text{grav}} = \lambda(-\log h) \text{ exactly} \\
\text{X5 (WKB + MPS)} & \xrightarrow{\text{unique s.c. flow}} & \text{leading coupling symbol, } \kappa \\
\text{X6 (KS class)} & \xrightarrow{\text{within-class = gauge}} & \text{coupling ordering mod (b)} \\
\text{X3 (direct sum)} & \xrightarrow{\text{GM per-sector}} & \text{assembly with } (\lambda_k) \\
\text{X7 (branch-uniform)} & \longrightarrow & \lambda_k \equiv \lambda \text{ (ambiguity (a))} \\
\text{residual of (i)–(v)} & \xrightarrow{\text{boundary triples}} & \text{extension data (ambiguity (c))}
\end{array}
$$

Joint use: clock closure = X1$\wedge$X4; matter closure = X4$\wedge$X5$\wedge$X6 (the over-determination); gravity closure = X2 alone; assembly = X3$\wedge$X7.

Independence and non-redundancy

- **X4 vs X7 (redundancy found and removed).** The old X7 ("restricts to the Part I constraint on finite-dim factors") was, at the level of the constraint *operator*, implied by X4: the flat classical-clock limit through finite-dimensional matter factors already fixes the Part I generator. The non-redundant content of X7 is only the branch-*uniform* normalization (and the boundary-data/Born-weight structure, which concerns the state space, not $\widehat{C}$, and is moved out of the axiom set). X7 has been restated accordingly.
- **X4 strict form would kill ambiguity (a).** If X4 demanded $c_k = 1$ (exact reduction), then $\lambda = 1$ and the "up to scale" clause of Thm. 3.1 is vacuous. The projective form is the consistent choice.
- **X6 conclusion-smuggling (referee attack sustained, axiom weakened).** The old X6 required reduction to "the modified Wheeler–DeWitt constraint of (U4)", whose statement contains $\alpha\widehat{\Sigma}_\delta$ — i.e. it assumed on the clock–geometry cross-sector precisely what X1+X4 are supposed to derive. The weakened X6 above is a pure universality-class membership statement (Kaimakkamis–Sil), makes no reference to Σ_δ, and demonstrably does independent work only in step (v) of Lemma 4.4.
- **X2 near-smuggling (assessed, kept, scoped).** X2 identifies the gravity generator's flow with the $\hat{\omega}$-modular flow; this is the main paper's thermal-time postulate, and relative to it X2 does the bulk of the gravity-sector work (Lemma 4.1 is then short). This is legitimate *relative* uniqueness (as Poincaré covariance is for P^μ), but the theorem must be advertised as "unique given thermal time," not as deriving thermal time. The non-degeneracy clause ($h \notin \mathbb{C}\mathbb{1}$) is required: without it X2 is vacuous on the II_1 sector and the conjunction X1–X7

loses gravity-sector selectivity entirely — this is the new failure mode introduced by the II_1 reclassification, and it is exactly why X2 cannot be stated w.r.t. the trace.

- **Mutual consistency / non-vacuous conjunction.** The de Sitter static-patch witness model of the main paper's (H6)/(U6) satisfies X1–X7: it is *natively* II_1 [10], so the reclassification improves rather than damages non-vacuity. $\widehat{C} = \widehat{\mathfrak{C}}_k$ itself satisfies all seven axioms on that model by construction.
- **X1's old form had a vacuous clause.** The commutator condition "$[\widehat{C}, \widehat{\Pi}_{\mathrm{c}} + \nVdash \otimes \nVdash] = 0$" contained an identity summand that commutes with everything; removed.

8 Open questions and triage

Provable now (write-up work only)

- Clock closure on $\mathcal{H}_{[E_\star]}$ (done in the main paper post-A29; import).
- Gravity rigidity Lemma 4.1 (elementary on II_1; Pedersen–Takesaki [7] + factor triviality of the relative commutant).
- Over-determination steps (i)–(iii); assembly + X7 scale-tying; the audit of Sec. 7.

Partially proven (external theorem in place, inverse/bridging argument to write)

- Step (iv): MPS-based WKB rigidity (forward well-posedness is [8]; inverse direction to write; scope limited to flat FLRW + massive scalar).
- Step (v) on minisuperspace ([9] is the input theorem; the "within-class = centralizer gauge" identification to write).

Genuinely open

- **O1** Lift of the Kaimakkamis–Sil universality classification beyond minisuperspace to the midisuperspace/QFT matter content of the main paper. *Principal conjectural input.*
- **O2** Step (vi): interior-trivial residuals are boundary-supported (boundary-triple formulation on DeWitt-metric minisuperspace).
- **O3** Partial collapse of the centralizer class (b) by X5 beyond leading WKB order.
- **O4** Clock closure for unbounded $\widehat{H}_{\mathrm{c}}$ without the band-limit.
- **O5** Relaxing X3 to a continuous field of II_1 factors over the branch index.
- **O6** Non-perturbative dynamical-horizon regime (II_∞ sectors of Rem. 4.3; residual (E3a) of the main paper).

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